

1. What is SQL?

SQL stands for Structured Query Language. It is a standard language used to retrieve and manage data in a relational database. We write SQL queries to perform operations such as retrieving, inserting, updating, deleting, and managing data.

2. Difference between Data and Database?

Data: Data is a raw fact which is used to describe an attribute of an entity.

Database: Database is a place/medium which is used to store a data in systematic and organized manner.

It has a 4 basic operation. Such as,

CREATE/UPDATE

READ/RETRIEVE

UPDATE/MODIFY

DELETE/DROP

This operation is referred as a CRUD operation.

3. Difference between DBMS and RDBMS?

DBMS	RDBMS
Database Management System	Relational Database Management System
The data stored in a file format	Data stored in table format
Difficult to retrieve data from a database	Query language is used to retrieve a data from a database.
Constraints are not used	Constraints are used such as primary key, foreign key, unique, null, check
Redundancy is high(duplicate values)	Low Redundancy.
Data integrity low	Data integrity high
Multiple users cannot access at a time	Multiple users can access the data at a time

4. What are datatypes and list the datatypes?

Datatype: Datatype are used to determine what kind of data we are going to store inside a cell or a particular memory allocation.

List of datatypes:

1. CHAR
2. VARCHAR/ VARCHAR2
3. DATE
4. NUMBER
5. Large Object
 - a. Character Large Object (CLOB)
 - b. Binary Large Object (BLOB)

5. What are constraints and list the constraints?

Constraints: Constraints are the set of assigned for a column.

It has 5 types. They are

1. UNIQUE
2. NOT NULL
3. CHECK
4. PRIMARY KEY
5. FOREIGN KEY

6. Difference between Primary key and foreign key

Features	Primary key	Foreign key
Purpose	Uniquely identifies records	Maintains referential integrity between tables

Definition	Primary key is a constraint which is used to identify a record uniquely from a table.	Foreign key is a constraint which is used to establish connection between the tables.
Number of keys per table	Only one Primary Key is allowed per table	Multiple Foreign Keys are possible in a single table
Combination of constraints	It is a combination of unique and not null constraints	It is not a combination of unique and not null constraints
Alternate Name	It is also known as Key attribute or Parent key	It is also known as a Referential Integrity Constraint.
Mandatory	It is not mandatory because it is a type of constraint. However, it is highly recommended for every table	It is not mandatory. It is used only when relationships between tables are required.
Ownership	It belongs to its own table	It is present in the child table, but its values originate from the parent table.
Requirement for Referencing	No dependency on other tables.	A Foreign Key must reference a column that is either a PRIMARY KEY or a UNIQUE key in the parent table.
Location	Defined in parent table	Defined in child table
References	Does not reference another table	References the Primary Key (or Unique Key) of another table
Purpose in Relationship	Parent identifier	Child reference
Constraint Keyword	PRIMARY KEY	FOREIGN KEY

7. What is Data Query Language?

Data Query language is used to retrieve a data from a database.

8. Explain in detail about projections?

Projection - Statement: It is a process of retrieving the data from database by selecting only the column is known as Projection.

In Projection all the records/values present in a particular column are by default selected.

Syntax:
SELECT */[DISTINCT] COL_NAME/EXPRESSION[ALIAS] FROM Table_Name;

DISTINCT Clause: Distinct clause is used to avoid repeated or duplicated value from the result table.

Distinct clause should be the first argument in a select clause.

Distinct compare combination of the column to consider a value as a repeated value.

Expression: Expression is a statement which is responsible to generate an output.

Expression is a combination of operand and operator

Operand: Operand are the values which we are pass (0,1,2,3...,n).

Operator: Operators are arithmetic symbol which perform operations on operand to generate an output (+, -, *, /, %).

Alias: Alias is used to assign alternate or a temporary name for column or expression in the result table.

We can assign alias by using the keywords.

Keyword: AS

We can assign alias as string (Multiple words), we have 2 formats They are

1. Annual_Salary
2. "Annual Salary"

Note: Keyword is optional (with or without)

Alias is also known as **Identifiers**.

Select sal*6 as Annual_Salary from EMP;

Annual_Salary/ "Annual Salary"/ Annual → Alias

AS → keyword

SAL*6 → Expression

9. Difference between Unique and Distinct clause

Feature	Unique	Distinct
Definition	Unique is a constraint which is used to avoid repeated or duplicated value from a column	Distinct is a keyword which is used to retrieve unique records from the query result.
Purpose	Maintains data integrity by preventing duplicate data from being stored.	Eliminates duplicate rows from the output of a SELECT statement.
Purpose	Constraint	SQL Keyword
When It Works	During INSERT and UPDATE operations	During data retrieval (SELECT)
NULL Values	Allows NULL values multiple times	NULL values appear only once in the result if they are duplicates
Effect on Stored Data	Prevents duplicate data from being stored in the table	Does not modify the stored data
Syntax	Defined while creating or altering a table	Used within a SELECT query
Example Syntax	Email VARCHAR2(100) UNIQUE	SELECT DISTINCT deptno FROM emp;

10. What is Selection:

It is a process of retrieving the data from database by selecting both the column and rows is known as selection.

Syntax:
SELECT */[DISTINCT] Column_name/Expression[Alias] FROM Table_name WHERE <Filter_Condition>;

11. List Operators in SQL?

1. Arithmetic Operator (+, -, /, *, %)
2. Concatenation Operator (||)
3. Comparison Operator (=, != or <>)
4. Relational Operator (>, <, >=, <=)
5. Logical Operator (AND, OR, NOT)
6. Special Operator (IN, NOT IN, IS NOT IN, BETWEEN, NOT BETWEEN, IS, IS NOT, LIKE, NOT LIKE)
7. Sub-Query Operator (ANY, ALL)

12. What are Logical Operators?

Logical operator is used to pass multiple conditions in WHERE clause.

In logical operators we have 3 types. They are,

1. AND
2. OR
3. NOT

13. What are Special Operators and list Special Operator?

Special operator is used to optimize or compress filter conditions in WHERE clause.

It has 8 types. They are,

1. IN
2. NOT IN
3. IS
4. NOT IS
5. BETWEEN
6. NOT BETWEEN
7. LIKE
8. NOT LIKE

14. Explain Escape Character?

It is used to remove special behaviour of wild character[%,_] and treat it as a normal character.

It should be written before the wild character.

Recommended escape characters are @, \$, &, #, /.

Syntax:

```
Column_name/Expression LIKE/NOT LIKE 'Pattern_to_Match'  
ESCAPE 'SPL_CHAR';
```

15. What is order by clause, it is syntax?

It is used to arrange a record either in ascending or descending order.

Syntax:

```
SELECT Group_BY_Expression/ Group_Function  
FROM Table_Name  
WHERE <Filter_Condition>  
GROUP BY Column_Name/Expression  
HAVING [<Group_Filter_Condition>]  
ORDER BY Column_Name [ASC]/DESC;
```

Syntax:

```
SELECT */[DISTINCT] Column_Name/Expression[Alias]  
FROM Table_name  
WHERE <Filter_Condition>
```

ORDER BY Column_Name [ASC]/DESC;

16. What are functions? What are the main components?

Function is list of instructions which is used to perform specific task.

There are 3 main components in function:

1. Function name
2. Number of arguments (Input)
3. Return Type

17. Explain Difference between Single row and multi row function?

Feature	Single row function	Multi row function
Definition	Single row functions process one row at a time and return one result for each row.	Multi row functions accept multiple rows as input, process them together, and return a single output.
Output	For single row function if we pass 'n' number of input it returns 'n' number of output	For multi row function if we pass 'n' number of input it returns a single output
Filter clause	It can used in both Where and Having Clause.	It is used only in Having clause
Example function name	LENGTH(),UPPER(), LOWER(), etc.	COUNT(), SUM(), AVG(), MIN(), MAX()
Example	select LENGTH(ename) from emp;	select MAX(sal) from emp;

18.

19. Explain Group by clause?

Group by clause is used to group the records.

Syntax:

```
SELETE Group_by_Expression/Group_Function
FROM Table_Name
[WHERE <Filter_Condtion>]
GROUP BY Column_Name/Expression;
```

20. Difference between Where clause and Having clause?

Feature	WHERE clause	HAVING clause
Definition	Where clause is used to filter the records individually (rows).	HAVING clause is used to filter groups of records.
Execute Order	Executes after the FROM clause and before the GROUP BY clause.	Executes after the GROUP BY clause.
Execution method	Evaluates the condition row by row.	Evaluates the condition group by group.
filter condition	Used to specify row-level filtering conditions.	Used to specify group-level filtering conditions.
Boolean condition	Evaluates a Boolean condition for each row.	Evaluates a Boolean condition for each group.

Pass Condition	Multiple conditions can be passed using logical operators (AND, OR, NOT).	Conditions involving aggregate (multi row) functions are generally specified here. Multiple conditions can also be passed using logical operators.
GROUP BY Dependency	Can be used with or without GROUP BY	Usually used with GROUP BY.
Example	WHERE SAL > 2000	HAVING AVG(SAL) > 2000

21. What is sub-Query:

A query is used inside another query is known as sub-query.

Working Principle of sub-Query:

Let us consider two query they are,

1. Outer Query
2. Inner Query/Sub Query

Inner Query starts the execution and produces an output.

The produced output is taken as an input to an Outer Query.

Outer query starts an execution and produces the result.

Hence, we can say that the outer query is depend on the inner query.

Why we use Sub-Query: Case 01 and Case 02 Definition:

Case 1: Whenever we have unknown present in the question we use subquery case-1 to find that unknown.

Case 2: Whenever the data to be selected and condition to be executed present in different tables we use subquery case 02.

22. Difference between the types of sub-query?

There are two types of sub-query:

1. Single row sub-query
2. Multi row sub-query

Feature	Single row Sub query	Multi-row sub query
Definition	Inner query/Sub query which returns single output is known as single row subquery	Inner query/sub query which returns multiple output is known as multi row subquery.
Operators	For single row subquery we can pass both normal operator(=) and special operator(IN)	For multirow subquery we should pass only special operator(IN), if we pass normal operator (=) it ends up in an error.
Example Query	Find employees working in the same department as SMITH.	Find employees working in departments located in NEW YORK.

23. Explain sub query operator with example?

Single row sub-query operator: For single row subquery we can pass both normal operator(=) and special operator(IN).

Multi row sub-query operator: For multirow subquery we should pass only special operator(IN), if we pass normal operator (=) it ends up in an error.

24. What is nested subquery giving an example?

A subquery written inside another subquery is known as nested subquery.

We can nest up to 255 subqueries.

25. What is correlated subquery?

A query written inside another query where both queries are inter-dependent is known as correlated subquery.

Working principle of correlated subquery:

Let us consider two queries they are,

1. Outer query
2. Inner query

In co-related subquery outer query starts the execution and gives us the partial output. The partial output of the outer query will be given as an input to the inner query.

After getting the input from the outer query inner query starts the execution and generates complete output.

The complete output is given as an input to the outer query and outer query starts the execution for the second time and generates the final output.

Therefore, outer query and inner query are inter-dependent (depend on each other). We call it as co-related sub-query.

26. What is Joins?

The process of retrieving data from multiple tables simultaneously is known as joins.

Why/ When we use joins: Whenever the attribute is selected from both the tables we use joins.

Types of Joins. (With definition and Syntax)

1. Cartesian join/ Cross join
2. Inner join/ Equi join
3. Outer Join
 - a. Left outer join
 - b. Right outer join
4. Natural join
5. Self-join

Cartesian join/cross join:

Definition: In cartesian join a whole table 1 will be merged with the first records of table 2 and it continues.

Column= Number_of_column_in_Table_1 + Number_of_column_in_Table_2;

Row= Number_of_Row_in_Table_1 * Number_of_Row_in_Table_2;

Let consider,

Number_of_column_in_Table_1 = 8

Number_of_column_in_Table_2 = 3

Column = 8+3 = 11 Columns

Number_of_Row_in_Table_1 = 14

Number_of_Row_in_Table_2 = 4

Row = 14*4 = 56 Rows

Syntax:
ANSI: SELECT column_name From Table_name_1 cross join Table_name_2; ORACLE: SELECT column_name From Table_name_1, Table_name_2;
Example: ANSI: Select * from emp cross join dept; ORACLE: Select * from emp, dept;

Inner join/ Equi join

Definition: Inner join is used to obtain only matching Records or A records which has a pair.

Join Condition Syntax:
Table_Name_1.Col_Name = Table_Name_2.Col_Name
Example: EMP.DEPTNO=DEPT.DEPTNO

Syntax:
ANSI: SELECT column_name FROM Table_Name_1 Inner Join Table_Name_2 ON <Join_Condition>; ORACLE: SELECT column_name FROM Table_Name_1, Table_Name_2 WHERE <Join_Condition>;
Example: SELECT * FROM emp Inner Join dept ON emp.deptno=dept.deptno; ORACLE: SELECT * FROM emp e, dept d WHERE e.deptno=d.deptno;

Outer join

Definition: It is used to obtain the unmatched records from the particular table along with the matched records.

3 Types:

1. Left Outer Join
2. Right Outer Join
3. Full Outer Join

Syntax: Left Outer Join
ANSI: SELECT Column_Name FROM Table_Name_1 Left Outer Join Table_Name_2 ON <Join_Condition>; ORACLE: SELECT Column_Name FROM Table_Name_1, Table_Name_2 WHERE Table_Name_1.Col_Name=Table_Name_2.Col_Name(+);
Example: ANSI: SELECT * FROM emp e Left Outer Join dept d ON e.deptno=d.deptno; ORACLE: SELECT * FROM emp e, dept d ON e.deptno=d.deptno(+);

Syntax: Right Outer Join
ANSI: SELECT Column_Name FROM Table_Name_1 Right Outer Join Table_Name_2 ON <Join_Condition>; ORACLE: SELECT Column_Name FROM Table_Name_1, Table_Name_2 WHERE Table_Name_1.Col_Name(+)=Table_Name_2.Col_Name;
Example: ANSI: SELECT * FROM emp e Right Outer Join dept d ON e.deptno=d.deptno; ORACLE: SELECT * FROM emp e, dept d WHERE e.deptno(+)=d.deptno;

Syntax: FULL OUTER JOIN → Not in Oracle only in ANSI
ANSI:

```
SELECT Column_Name
FROM Table_Name_1 Full [Outer] Join Table_Name_2
ON <Join_Condition>;
```

Example:
ANSI:
SELECT *
FROM emp e Full Outer Join dept d
ON e.deptno=d.deptno;

Natural join

Definition: It behaves as “Inner Join” if there is a relation between the given tables, else it behaves as “Cross Join”.

Syntax:

ANSI:
Select Column_Name
From table_name_1 NATURAL JOIN table_name_2;

Example:
Select *
From emp NATURAL JOIN dept;

Behave as:
Select *
From emp INNER JOIN dept;
Select *
From emp CROSS JOIN salgrade;

Self-join

Definition: Joining the table by itself is known as self-join.

Syntax:

ANSI:
SELECT Column_Name
FROM Table_Name T1 Join Table_Name T2
ON <Join_Condition>;
ORACLE:
SELECT Column_Name
FROM Table_Name T1, Table_Name T2
WHERE <Join_Condition>;

Example:
ANSI:
SELECT *
FROM emp e1 JOIN emp e2
ON e1.mgr = e2.empno;
ORACLE:
SELECT *
FROM emp e1, emp e2
WHERE e1.mgr = e2.empno;

27. Explain Data Definition Language (DDL)?

Called:

Alias → Identifiers

AS → keyword

DISTINCT → Keyword